Docker

install:

```
curl -fsSL get.docker.com -o get-docker.sh
sudo sh get-docker.sh

sudo apt-get install docker-compose-plugin
sudo apt install docker-compose
```

show images:

```
sudo docker images
```

remove image:

```
sudo docker rmi <id>
```

show running containers:

```
sudo docker ps
```

prune before removal

```
sudo docker system prune
```

build container

```
docker build -t alexcontainer .
. = Dockerfile located in directory command run from
```

```
-t = name the container
```

start container

```
docker run -d --restart unless-stopped container_id
```

map port

```
docker run --name webserver -p 8080:80 -d alexcontainer --
restart always
--name = give memorable name
-p = map port i.e. 8080 from host to 80 in container
-d = daemon and run container called alexcontainer
```

execute commands in container

```
get container id
docker exec -it d2d4a89aaee9 ip addr
docker exec -it <containerid> bash
docker exec -it <containerid> bash
remember you can shorten IDs to 2 or 3 characters!
```

Docker-compose

docker compose up

```
sudo docker-compose up -d
-d = run as daemon
```

auto-restart

```
add to compose file:

services:
        restart: always
```

Create network:

```
docker network create -d macvlan --subnet=192.168.1.0/24 --
gateway=192.168.1.1 -o parent=eth0 custom_network
```

Create VM with IP 192.168.1.22 on this network

```
docker run --network=custom_network --ip=192.168.1.22 -d
your-image
```

View networks:

```
docker network ls
```

Inspect network:

```
docker network inspect [NETWORK_NAME_OR_ID]
```

Delete network:

```
docker network rm [NETWORK_NAME_OR_ID]
```

Pull from specific registry:

```
docker pull myregistry.com/myimage
```